

Number of Prime Ministers Per Province/Territory Circuit

CP220 Project Phase II

Lukas Palubiski

169037714

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Truth tables and Karnaugh Maps

Here is the full truth table of all the inputs and outputs:

x_1	x_2	x_3	x_4		y_1	y_2	y_3	y_4
0	0	0	0		0	0	1	0
0	0	0	1		0	0	0	1
0	0	1	0		0	0	1	0
0	0	1	1		0	0	1	0
0	1	0	0		0	0	0	0
0	1	0	1		0	0	0	1
0	1	1	0		0	0	1	0
0	1	1	1		0	0	0	0
1	0	0	0		1	1	1	1
1	0	0	1		0	0	0	1
1	0	1	0		1	0	1	1
1	0	1	1		0	0	1	0
1	1	0	0		0	0	0	0
1	1	0	1		0	0	0	0
1	1	1	0		0	0	0	0
1	1	1	1		0	0	0	0

Y_1 Truth Table:

x_1	x_2	x_3	x_4	y_1
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y_1 Karnaugh Map

$x_1, 2, 3, 4$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

Solution: X_1, X_2', X_4'

Y₂ Truth Table

x ₁	x ₂	x ₃	x ₄	y ₂
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y₂ Karnaugh Map

X _{1, 2, 3, 4}	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	0	0	0

Solution: $X_1, X_2'X_3', X_4'$

Y₃ Truth Table

X ₁	X ₂	X ₃	X ₄	Y ₃
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y₃ Karnaugh Map

X _{1, 2, 3, 4}	00	01	11	10
00	1	0	1	1
01	0	0	0	1
11	0	0	0	0
10	1	0	1	1

Solution: $X_2', X_4' + X_2'X_3 + X_1', X_3, X_4'$

Y₄ Truth Table

x ₁	x ₂	x ₃	x ₄	y ₄
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y₄ Karnaugh Map

X _{1, 2, 3, 4}	00	01	11	10
00	0	1	0	0
01	0	1	0	0
11	0	0	0	0
10	1	1	0	1

Solution: $X_1', X_3', X_4 + X_1, X_2', X_4' + X_2' X_3', X_4$

All Code for Maxima Testing:

Y1: x1 and (not x2) and (not x4);

Y1, x1=false, x2=false, x3=false, x4=false;
Y1, x1=false, x2=false, x3=false, x4=true;
Y1, x1=false, x2=false, x3=true, x4=false;
Y1, x1=false, x2=false, x3=true, x4=true;
Y1, x1=false, x2=true, x3=false, x4=false;
Y1, x1=false, x2=true, x3=false, x4=true;
Y1, x1=false, x2=true, x3=true, x4=false;
Y1, x1=false, x2=true, x3=true, x4=true;
Y1, x1=true, x2=false, x3=false, x4=false;
Y1, x1=true, x2=false, x3=false, x4=true;
Y1, x1=true, x2=false, x3=true, x4=false;
Y1, x1=true, x2=false, x3=true, x4=true;
Y1, x1=true, x2=true, x3=false, x4=false;
Y1, x1=true, x2=true, x3=false, x4=true;
Y1, x1=true, x2=true, x3=true, x4=false;
Y1, x1=true, x2=true, x3=true, x4=true;

Y2:(x1 and not x2 and not x3 and not x4);

Y2, x1=false, x2=false, x3=false, x4=false;
Y2, x1=false, x2=false, x3=false, x4=true;
Y2, x1=false, x2=false, x3=true, x4=false;
Y2, x1=false, x2=false, x3=true, x4=true;
Y2, x1=false, x2=true, x3=false, x4=false;
Y2, x1=false, x2=true, x3=false, x4=true;
Y2, x1=false, x2=true, x3=true, x4=false;
Y2, x1=false, x2=true, x3=true, x4=true;
Y2, x1=true, x2=false, x3=false, x4=false;
Y2, x1=true, x2=false, x3=false, x4=true;
Y2, x1=true, x2=false, x3=true, x4=false;
Y2, x1=true, x2=false, x3=true, x4=true;
Y2, x1=true, x2=true, x3=false, x4=false;
Y2, x1=true, x2=true, x3=false, x4=true;
Y2, x1=true, x2=true, x3=true, x4=false;
Y2, x1=true, x2=true, x3=true, x4=true;

Y3: (not x2 and not x4) or (not x2 and x3) or (not x1 and x3 and not x4);

Y3, x1=false, x2=false, x3=false, x4=false;
Y3, x1=false, x2=false, x3=false, x4=true;
Y3, x1=false, x2=false, x3=true, x4=false;
Y3, x1=false, x2=false, x3=true, x4=true;
Y3, x1=false, x2=true, x3=false, x4=false;
Y3, x1=false, x2=true, x3=false, x4=true;
Y3, x1=false, x2=true, x3=true, x4=false;
Y3, x1=false, x2=true, x3=true, x4=true;
Y3, x1=true, x2=false, x3=false, x4=false;
Y3, x1=true, x2=false, x3=false, x4=true;
Y3, x1=true, x2=false, x3=true, x4=false;
Y3, x1=true, x2=false, x3=true, x4=true;
Y3, x1=true, x2=true, x3=false, x4=false;
Y3, x1=true, x2=true, x3=false, x4=true;
Y3, x1=true, x2=true, x3=true, x4=false;
Y3, x1=true, x2=true, x3=true, x4=true;

Y4:(not x1 and not x3 and x4) or (x1 and not x2 and not x4) or (not x2 and not x3 and x4);

Y4, x1=false, x2=false, x3=false, x4=false;
Y4, x1=false, x2=false, x3=false, x4=true;
Y4, x1=false, x2=false, x3=true, x4=false;
Y4, x1=false, x2=false, x3=true, x4=true;
Y4, x1=false, x2=true, x3=false, x4=false;
Y4, x1=false, x2=true, x3=false, x4=true;
Y4, x1=false, x2=true, x3=true, x4=false;
Y4, x1=false, x2=true, x3=true, x4=true;
Y4, x1=true, x2=false, x3=false, x4=false;
Y4, x1=true, x2=false, x3=false, x4=true;
Y4, x1=true, x2=false, x3=true, x4=false;
Y4, x1=true, x2=false, x3=true, x4=true;
Y4, x1=true, x2=true, x3=false, x4=false;
Y4, x1=true, x2=true, x3=false, x4=true;
Y4, x1=true, x2=true, x3=true, x4=false;
Y4, x1=true, x2=true, x3=true, x4=true;

All Outputs for Testing

Y1 Testing

```
(%i1)      Y1: x1 and (not x2) and (not x4);

(%o1)       $x_1 \wedge \neg x_2 \wedge \neg x_4$ 

(%i2)      Y1, x1=false, x2=false, x3=false, x4=false;

(%o2)      false

(%i3)      Y1, x1=false, x2=false, x3=false, x4=true;

(%o3)      false

(%i4)      Y1, x1=false, x2=false, x3=true, x4=false;

(%o4)      false

(%i5)      Y1, x1=false, x2=false, x3=true, x4=true;

(%o5)      false

(%i6)      Y1, x1=false, x2=true, x3=false, x4=false;

(%o6)      false

(%i7)      Y1, x1=false, x2=true, x3=false, x4=true;

(%o7)      false
```

(%i8) Y1, x1=false, x2=true, x3=true, x4=false;

(%o8) false

(%i9) Y1, x1=false, x2=true, x3=true, x4=true;

(%o9) false

(%i10) Y1, x1=true, x2=false, x3=false, x4=false;

(%o10) true

(%i11) Y1, x1=true, x2=false, x3=false, x4=true;

(%o11) false

(%i12) Y1, x1=true, x2=false, x3=true, x4=false;

(%o12) true

(%i13) Y1, x1=true, x2=false, x3=true, x4=true;

(%o13) false

(%i14) Y1, x1=true, x2=true, x3=false, x4=false;

(%o14) false

(%i15) Y1, x1=true, x2=true, x3=false, x4=true;

(%o15) false

(%i16) Y1, x1=true, x2=true, x3=true, x4=false;

(%o16) false

(%i17) Y1, x1=true, x2=true, x3=true, x4=true;

(%o17) False

Y2 Testing

(%i18) Y2: (x1 and not x2 and not x3 and not x4);

(%o18) $x_1 \wedge \neg x_2 \wedge \neg x_3 \wedge \neg x_4$

(%i19) Y2, x1=false, x2=false, x3=false, x4=false;

(%o19) false

(%i20) Y2, x1=false, x2=false, x3=false, x4=true;

(%o20) false

(%i21) Y2, x1=false, x2=false, x3=true, x4=false;

(%o21) false

(%i22) Y2, x1=false, x2=false, x3=true, x4=true;

(%o22) false

(%i23) Y2, x1=false, x2=true, x3=false, x4=false;

(%o23) false

(%i24) Y2, x1=false, x2=true, x3=false, x4=true;

(%o24) false

(%i25) Y2, x1=false, x2=true, x3=true, x4=false;

(%o25) false

(%i26) Y2, x1=false, x2=true, x3=true, x4=true;

(%o26) false

(%i27) Y2, x1=true, x2=false, x3=false, x4=false;

(%o27) true

(%i28) Y2, x1=true, x2=false, x3=false, x4=true;

(%o28) false

(%i29) Y2, x1=true, x2=false, x3=true, x4=false;

(%o29) false

(%i30) Y2, x1=true, x2=false, x3=true, x4=true;

(%o30) false

(%i31) Y2, x1=true, x2=true, x3=false, x4=false;

(%o31) false

(%i32) Y2, x1=true, x2=true, x3=false, x4=true;

(%o32) false

(%i33) Y2, x1=true, x2=true, x3=true, x4=false;

(%o33) false

(%i34) Y2, x1=true, x2=true, x3=true, x4=true;

(%o34) False

Y3 Testing

%i35) Y3: (not x2 and not x4) or (not x2 and x3) or (not x1 and x3 and not x4);

(%o35) $\neg x_2 \wedge \neg x_4 \vee \neg x_2 \wedge x_3 \vee \neg x_1 \wedge x_3 \wedge \neg x_4$

(%i36) Y3, x1=false, x2=false, x3=false, x4=false;

(%o36) true

(%i37) Y3, x1=false, x2=false, x3=false, x4=true;

(%o37) false

(%i38) Y3, x1=false, x2=false, x3=true, x4=false;

(%o38) true

(%i39) Y3, x1=false, x2=false, x3=true, x4=true;

(%o39) true

(%i40) Y3, x1=false, x2=true, x3=false, x4=false;

(%o40) false

(%i41) Y3, x1=false, x2=true, x3=false, x4=true;

(%o41) false

(%i42) Y3, x1=false, x2=true, x3=true, x4=false;

(%o42) true

(%i43) Y3, x1=false, x2=true, x3=true, x4=true;

(%o43) false

(%i44) Y3, x1=true, x2=false, x3=false, x4=false;

(%o44) true

(%i45) Y3, x1=true, x2=false, x3=false, x4=true;

(%o45) false

(%i46) Y3, x1=true, x2=false, x3=true, x4=false;

(%o46) true

(%i47) Y3, x1=true, x2=false, x3=true, x4=true;

(%o47) true

(%i48) Y3, x1=true, x2=true, x3=false, x4=false;

(%o48) false

(%i49) Y3, x1=true, x2=true, x3=false, x4=true;

(%o49) false

(%i50) Y3, x1=true, x2=true, x3=true, x4=false;

(%o50) false

(%i51) Y3, x1=true, x2=true, x3=true, x4=true;

(%o51) false

Y4 Testing

(%i5
2) Y4: (not x1 and not x3 and x4) or (x1 and not x2 and not x4) or (not x2 and not
x3 and x4);

(%o52 $\neg$
) $x_1 \wedge \neg x_3 \wedge x_4 \vee x_1 \wedge \neg x_2 \wedge \neg x_4 \vee \neg x_2 \wedge \neg x_3 \wedge x_4$

(%i5 Y4, x1=false, x2=false, x3=false, x4=false;
3)

(%o53 false
)

```
(%i5  Y4, x1=false, x2=false, x3=false, x4=true;  
4)
```

```
(%o54  true  
)
```

```
(%i5  Y4, x1=false, x2=false, x3=true, x4=false;  
5)
```

```
(%o55  false  
)
```

```
(%i5  Y4, x1=false, x2=false, x3=true, x4=true;  
6)
```

```
(%o56  false  
)
```

```
(%i5  Y4, x1=false, x2=true, x3=false, x4=false;  
7)
```

```
(%o57  false  
)
```

```
(%i5  Y4, x1=false, x2=true, x3=false, x4=true;  
8)
```

```
(%o58  true  
)
```

```
(%i5  Y4, x1=false, x2=true, x3=true, x4=false;  
9)
```

```
(%o59  false  
)
```

```
(%i6  Y4, x1=false, x2=true, x3=true, x4=true;  
0)
```

```
(%o60  false  
)
```

```
(%i6  Y4, x1=true, x2=false, x3=false, x4=false;  
1)
```

(%o61 true
)

(%i6 Y4, x1=true, x2=false, x3=false, x4=true;
2)

(%o62 true
)

(%i6 Y4, x1=true, x2=false, x3=true, x4=false;
3)

(%o63 true
)

(%i6 Y4, x1=true, x2=false, x3=true, x4=true;
4)

(%o64 false
)

(%i6 Y4, x1=true, x2=true, x3=false, x4=false;
5)

(%o65 false
)

(%i6 Y4, x1=true, x2=true, x3=false, x4=true;
6)

(%o66 false
)

(%i6 Y4, x1=true, x2=true, x3=true, x4=false;
7)

(%o67 false
)

(%i6 Y4, x1=true, x2=true, x3=true, x4=true;
8)

(%o68 false
)

Summary

As you can see from the truth tables, Karnaugh maps, as well as the Maxima outputs, everything is working as expected, and all equations pass all tests. I will attach the checklist here at the bottom.

PC/CP220 Project Phase II Checklist (3.01)

A. General

1. Professionally presented
Neat, etc. _____
2. Properly identified
(eg. name, id) _____
3. On time
at beginning of lab with checklist _____
4. Good grammar
(eg. complete sentences where required) _____
5. Correct spelling _____

B. Content

1. *All outputs analyzed
(truth table)* _____
2. *Karnaugh maps (or equations)
match truth table* _____
3. *Equations tested
(Maple, Maxima, etc.)* _____
4. *Summary statement
(all equations pass all tests)* _____
5. *Previous phase included
(updated if necessary)* _____